

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (Mechanical) (Credit System) Degree Examinations

January 2023

19MEE54 – COMPUTER INTEGRATED MANUFACTURING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Briefly outline several plant layouts utilized for various types of production.	10	L*1	1	1
	b) Explain different types of automation relative to production quantity and product variety.	10	L2	2	1
2.	a) Illustrate various types of production systems that are categorized according to the quantity of products produced.	10	L1	1	1
	b) Describe the several causes for automating the production industry.	10	L2	2	1
3.	a) Explain the components of a typical manufacturing firm's information-processing cycle.	10	L1	1	1
	b) Using a simple sketch, demonstrate several possible layouts of an automated production line.	10	L2	2	1
Unit – II					
4.	a) Outline various types of automated assembly systems based on physical configuration.	10	L1	3	1
	b) Explain briefly the contact and non-contact inspection techniques.	10	L2	4	1
5.	a) Demonstrate the components of a parts delivery system that are utilized at an automated assembly workstation.	10	L1	3	1
	b) Explain several physical setups for accomplishing probe motion in a coordinate measuring machine.	10	L2	4	1
6.	a) What are the suggestions and approaches that may be used in product design to make automated assembly systems easier to use?	10	L1	3	1
	b) For a single station assembly station explain how ideal cycle time, machine cycle time and line efficiency are calculated.	10	L2	4	1
Unit – III					
7.	a) Define the AS/RS system as well as the goals for automating a company's storage operations.	10	L1	5	1
	b) Classify the carousel system and describe the several types of carousel storage controls.	10	L2	5	1
8.	a) Describe the several types of AS/RS systems.	10	L1	5	1
	b) Outline some useful applications for carousel storage systems.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
